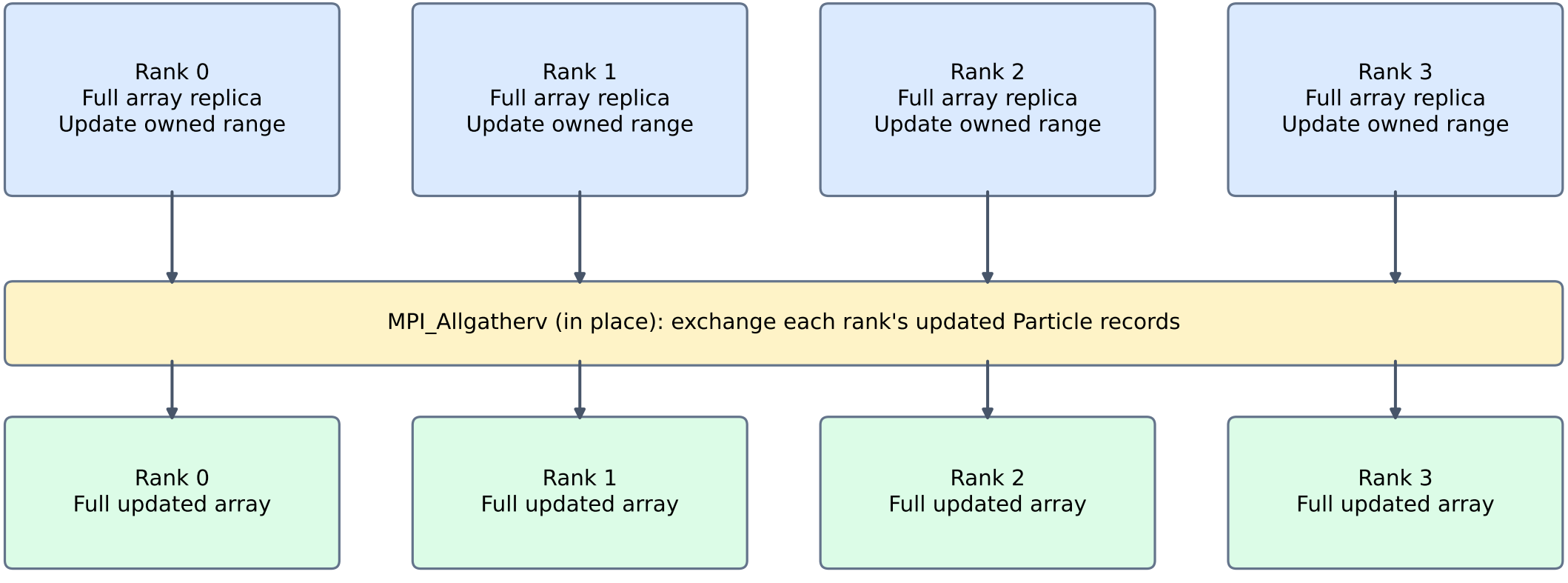
MPI: local work, globally replicated states

Rank 0 loads data and broadcasts it once. Each timestep uses MPI_Allgatherv to exchange updated particles.



Counts and displacements handle uneven partitions. Repeat force, position and exchange steps for every iteration.